

run_sjsdm_predictor_structure_folds.R

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Contents

run_sjsdm_predictor_structure_folds 1

Index 3

run_sjsdm_predictor_structure_folds

Run One sjSDM Predictor Structure Across Selected Folds

Description

Applies one controlled predictor structure to deterministic selected-candidate folds and returns labelled predictions and diagnostics.

Usage

```
run_sjsdm_predictor_structure_folds(  
  predictor_structure = NULL,  
  data_assignments = NULL,  
  data_selected_candidate = NULL,  
  data_community_matrix = NULL,  
  data_abiotic_wide = NULL,  
  data_coords_projected = NULL,  
  data_sample_ids = NULL,  
  config_model_fitting = NULL,  
  config_data_processing = NULL,  
  model_formula = NULL,  
  device = "gpu",  
  seed = 900723L,  
  prepare_fold_function = prepare_sjsdm_cross_validation_fold,  
  fit_candidate_function = fit_sjsdm_cross_validation_candidate,  
  runner_function = run_sjsdm_selected_candidate_folds,  
  predict_function = predict_sjsdm_probability_matrix  
)
```

Arguments

<code>predictor_structure</code>	Structure passed to <code>[build_sjsdm_predictor_comparison_structure()]</code> .
<code>data_assignments</code> , <code>data_selected_candidate</code> , <code>data_community_matrix</code>	Cross-validation assignments, selected regularization, and response matrix.
<code>data_abiotic_wide</code> , <code>data_coords_projected</code> , <code>data_sample_ids</code>	Abiotic predictors, projected coordinates, and aligned sample metadata.
<code>config_model_fitting</code> , <code>config_data_processing</code>	Model-fitting and data-processing configurations used by fold preparation.
<code>model_formula</code>	Full abiotic formula used when abiotic predictors are enabled.
<code>device</code>	Fitting device passed to <code>[fit_sjsdm_regularization_candidate()]</code> .
<code>seed</code>	Base selected-fold seed passed to <code>[run_sjsdm_selected_candidate_folds()]</code> .
<code>prepare_fold_function</code> , <code>fit_candidate_function</code> , <code>runner_function</code>	Injectable functions for fold preparation, candidate fitting, and selected fold orchestration.
<code>predict_function</code>	Injectable held-out prediction function.

Details

Regularization and assignments are fixed by the supplied reference inputs. The helper changes predictor inclusion only. Intercept-only and spatial-only structures use an environmental intercept because sjSDM requires an environmental component. The no-associations structure retains abiotic and spatial predictors while restricting the biotic covariance to its diagonal.

Value

Named list containing labelled ‘data_predictions’ and ‘data_diagnostics’.

Examples

```
## Not run:
run_sjsdm_predictor_structure_folds(
  predictor_structure = "abiotic_only",
  data_assignments = data_assignments,
  data_selected_candidate = data_selected_candidate,
  data_community_matrix = data_community_matrix,
  data_abiotic_wide = data_abiotic_wide,
  data_coords_projected = data_coords_projected,
  data_sample_ids = data_sample_ids,
  config_model_fitting = config_model_fitting,
  config_data_processing = config_data_processing,
  model_formula = model_formula
)

## End(Not run)
```

Index

`run_sjsdm_predictor_structure_folds`, [1](#)